Build a complete prediction

LEARNING TARGET I can connect forces, velocity change, and final position.

A · KEY VOCABULARY

The chain	Net force → acceleration → velocity → position.
Constant net force	Gives constant acceleration: a straight v–t line.
Zero net force	Velocity stays the same; position still changes.
Range	A central value with limits. Not a guarantee.

B · KEY EQUATIONS

$$a = F_{net} \div m$$
$$v_f = v_i + a \times t$$
$$\Delta x = \tfrac{1}{2}(v_i + v_f) \times t$$
$$x_f = x_i + \Delta x$$

`N, kg` force, mass

`m/s², m/s, m` a, v, x

C · COLOR IT · ONE CHAIN, FOUR LINKS

2 kg cart, from rest at 1 m, for 2 s.

1 NET FORCE
`6 − 2 = 4 N`
pull minus friction

→

2 ACCEL.
`4 ÷ 2 = 2 m/s²`
$F_{net} \div m$

→

3 VELOCITY
`0 + 2×2 = 4 m/s`
gain 2 m/s each second

→

4 POSITION
`1 + 4 = 5 m`
Δx = 2 m/s avg × 2 s

Report the range: **`5 ± 1 m → 4 to 6 m`** – the central value is not guaranteed.

☐ **1 · ORANGE** Force

☐ **2 · GREEN** Acceleration (speeding up)

☐ **3 · PURPLE** Velocity

☐ **4 · BLUE** Position

D · WORKED EXAMPLE

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement.

GIVEN	EQUATION	WORK	ANSWER
F_{net} = 0 N v = +2 m/s t = 3 s	a = 0 → v constant Δx = v × t	v stays +2 m/s Δx = 2 × 3	**+2 m/s, +6 m**

E · THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else